EFU-ENERGY STANDARD v1.0 - FINAL INTEGRATED VERSION

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Comprehensive Framework for Human-Centered Resource Governance

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EXECUTIVE SUMMARY – REVISED (publication-ready)

The EFU (Equivalent Flux Unit) system humanizes resource accounting by anchoring all measurements to human metabolic flows. Conventional technical units (joules, kilowatt-hours, tonnes) often obscure the biological and ecological costs underlying technological civilization.

This document serves as a reference standard for auditors, policymakers and infrastructure operators who require a human-centred energy accounting framework.

Why This Matters

- Translates abstract energy flows into human-relatable terms
- Enables governance without technical barriers (e.g. “equivalent to X people”)
- Reveals the true resource costs of digital infrastructure (AI, cloud, blockchain)
- Provides a unified accounting framework across energy, carbon, water and nutrient systems

EFU-E (Energy) Baseline

The EFU-E (Energy) baseline represents **human metabolic turnover**: the continuous energy flow required to sustain an average adult’s life and basic activities.

Key Reference Value

- **1 EFU-E = 3,154 MJ per person per year** (average adult total energy metabolism)

Alternative Expressions

- 1 EFU-E = 100 W per person (continuous metabolic power)

- 1 EFU-E = 2,000 kcal per day per person (dietary energy equivalent)
- 1 EFU-E = 876 kWh per person per year (electrical equivalent)

What This Standard Enables

- Tracking energy consumption in human-relatable units
- Quantifying per-capita energy intensity across sectors
- Comparing renewable and fossil energy transitions
- Assessing energy-efficiency improvements in metabolic terms
- Governing the resource costs of digital infrastructure (**AI, cloud, data centres**)

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1. SCIENTIFIC FOUNDATION — REVISED (publication-ready)

1.1 Human Energy Metabolism

Basal Metabolic Rate (BMR)

The human body continuously consumes energy to sustain life through the following processes:

- Cellular respiration and ATP production
- Protein synthesis and tissue maintenance
- Cardiovascular and respiratory function
- Neural activity and thermoregulation

Average Adult Reference Values

- Basal metabolic rate (BMR): approximately **70–85 W** (at rest)
- Daily average metabolic power (including normal activity): approximately **100 W continuous**
- Daily energy expenditure: **100 W × 86,400 s = 8.64 MJ ≈ 2,064 kcal**
- Annual energy expenditure: **100 W × 31,536,000 s = 3,154 MJ**

EFU-E Reference Standard

1 EFU-E = 3,154 MJ per person per year
= 100 W continuous metabolic power
= 2,000 kcal per day (rounded dietary reference)
= 876 kWh per year

Human Energy Balance

Energy intake (food)

= Basal metabolism + Physical activity + Thermogenesis + Growth/storage

For a metabolically stable adult:

Energy intake ≈ Energy expenditure
≈ 2,000 kcal/day ≈ 1 EFU-E per year

1.2 Global Energy Context

Human Metabolic Energy (Global Scale)

Global population: ~8 billion people
Total metabolic energy: $8 \times 10^9 \times 3{,}154$ MJ = 25.2 EJ/year
Equivalent continuous power: $8 \times 10^9 \times 100$ W = 800 GW

This represents the **baseline biological energy throughput of humanity**.

Anthropogenic Energy Consumption (circa 2024)

Global primary energy consumption: ~600 EJ/year
Per-capita energy use:
600 EJ / 8 billion ≈ 75,000 MJ/person/year
EFU-E equivalent:
75,000 / 3,154 ≈ 23.8 EFU-E per person

Interpretation

The average person currently relies on approximately **24 times their own metabolic energy** through industrial, transport, residential and digital energy systems.

Regional Variation in Energy Use

Region	Energy Use (MJ/cap/year)	EFU-E/person
North America	250,000	79.3
Europe	125,000	39.6
China	95,000	30.1
Global average	75,000	23.8
India	25,000	7.9
Sub-Saharan Africa	12,000	3.8

Energy Poverty Threshold

Minimum modern energy access: ~10,000 MJ/person/year
≈ 3.2 EFU-E per person

This level corresponds to basic lighting, cooking, essential appliances and minimal transport access.

Estimated population below this threshold: approximately **1 billion people**.

1.3 Energy Quality and Exergy

Not All Energy Is Equal

Energy exists in multiple forms with differing capacities to perform useful work, referred to as **exergy**.

High-Quality (High-Exergy) Energy

- Electricity: ~100% available work
- Mechanical work: ~100% available work
- High-temperature heat: ~60–80% available work
- Chemical energy (fuels): ~40–90% available work

Low-Quality (Low-Exergy) Energy

- Low-temperature heat (<100 °C): ~10–30% available work
- Ambient waste heat: ~0% available work

EFU-E Accounting Convention

By default, EFU-E measures GROSS (primary) energy,
not exergy or final useful work.

Rationale

- Primary energy is directly measurable and comparable
- Exergy and efficiency can be specified separately when relevant

Illustrative Example

Coal-fired power plant:
Input: 10 EFU-E (fuel)
Output: 3.5 EFU-E (electricity)

Efficiency: 35%
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2. STRATEGIC FRAMEWORK FOR GOVERNANCE — REVISED (publication-ready)

2.1 Primary vs. Final Energy Accounting

Critical Distinction for Audit Integrity

The distinction between **primary energy** and **final energy** is foundational to the intellectual and methodological integrity of the EFU-E system. Without this distinction, the resource and environmental impacts of fossil energy systems are systematically underestimated.

Fossil Energy Sources — Account for PRIMARY Energy

Rationale

Primary energy accounting captures the full rate of resource depletion, including extraction losses and thermodynamic conversion losses.

Illustrative Example: Coal-Fired Power Plant

Coal input (primary energy): 100 MJ
Conversion losses (≈65%): 65 MJ (waste heat)
Electricity delivered (final): 35 MJ

EFU-E Accounting

Primary energy: 100 MJ / 3,154 = 31.7 EFU-E → REPORT
Final energy: 35 MJ / 3,154 = 11.1 EFU-E → supplementary only

Why the 65 MJ loss matters

- Thermodynamic irreversibility (see EFU-Entropy Standard)
- Physical depletion of extracted fuel resources
- Environmental heat pollution (thermal discharge)

Ignoring these losses creates a distorted picture of system-level efficiency.

Renewable Energy Sources — Account for FINAL Energy

Rationale

Renewable energy systems do not consume fuel stocks; they convert freely available natural flows (wind, solar radiation) through infrastructure.

Illustrative Example: Wind Turbine

Primary energy input: N/A (free, unbounded natural flow)
Electricity generated: 1 MJ (final energy)

EFU-E Accounting

Final energy: 1 MJ / 3,154 = 0.317 EFU-E → REPORT
Primary energy: not applicable (or attributed to solar flux for theoretical analysis only)

Only delivered energy is relevant for accounting purposes, as no finite resource is depleted.

Nuclear Energy — Special Case

Nuclear fission releases energy through mass–energy conversion ($E = mc^2$).

EFU-E Convention

- Count **FINAL energy**, analogously to renewables
- Treat uranium mining, enrichment and construction energy as **separate embodied energy flows**

Rationale

Nuclear energy does not involve combustion-based fuel depletion and therefore constitutes a distinct physical category.

Why This Distinction Prevents Greenwashing

Without primary vs. final separation

"Coal plant efficiency: 40%"
"Wind power efficiency: 100%"

This framing creates a **false equivalence**.

With EFU-E accounting

Coal (40% efficiency):
  100 MJ fuel → 40 MJ electricity
  PRIMARY: 31.7 EFU-E
  FINAL: 12.7 EFU-E
  REPORT: 31.7 EFU-E

Wind:

40 MJ electricity
FINAL: 12.7 EFU-E
REPORT: 12.7 EFU-E

For the same delivered electricity, coal requires **~2.5× higher EFU-E**, reflecting its true resource cost.

Standardized Reporting Formats

Option A: Primary Energy Equivalent (RECOMMENDED)
- Account energy at the source
- Fossil fuels: include conversion losses
- Grid electricity: apply 2.86× multiplier (≈35% generation efficiency)
- Renewables: count final delivered energy

Option B: Final Energy Only
- Account energy as delivered
- Specify energy source mix separately
- Limitation: understates fossil resource depletion

EFU-E Requirement

All reports **must explicitly state** which accounting option is used.
For policy, auditing and cross-system comparison, **Option A is mandatory**.

2.2 Personal Energy Allowances (PEA)

Concept

Just as carbon budgets constrain emissions, **energy budgets must constrain total energy throughput** to remain within planetary boundaries.

1.5 °C-Compatible Energy Pathway

Current global energy use:
600 EJ/year ≈ 23.8 EFU-E per person (highly unequal)

Projected 2050 pathway:
Population: ~9.7 billion
Total energy: ~500 EJ/year
Per capita: 51.5 TJ/person/year ≈ 16.3 EFU-E
Renewable share: >90%

Equity Principle

- Global convergence toward **15–20 EFU-E per person**

- High-consumption societies (>40 EFU-E): **absolute reductions required**
- Low-consumption societies (<5 EFU-E): **energy access must increase**
- Middle-range societies: efficiency and structural optimization

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Indicative National Energy Budgets

(EFU-E per person per year)

Country group	Current	2030 target	2050 target
High income	60	45	25
Upper-middle	25	22	18
Lower-middle	10	13	16
Low income	4	8	15
Global average	23.8	20	16.3

2.3 Energy Intensity Benchmarking

GDP-Based Energy Intensity

(EFU-E per 1,000 USD GDP)

Global average (2024): ~5.0 EFU-E / $1,000 GDP

Advanced economies: ~3.5 EFU-E
Emerging economies: ~7.5 EFU-E

Long-Term Target (2050)

< 2.0 EFU-E per $1,000 GDP
≈ 60% reduction in energy intensity

Decarbonisation and Efficiency Pathways

1. Structural economic shift (services > heavy manufacturing)
2. Electrification of end uses (higher conversion efficiency)
3. Renewable energy deployment (elimination of combustion losses)
4. End-use efficiency (buildings, motors, lighting, transport)

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3. CONNECTION TO GAIA-X AND EU DIGITAL SOVEREIGNTY

3.1 Why EFU-E Matters for Gaia-X

Gaia-X is the European initiative for federated, sovereign cloud infrastructure, defining frameworks for data governance, interoperability, and trust. While these specifications effectively address **legal and informational dimensions** of digital sovereignty, they currently lack **systematic accounting of physical resource use**, particularly energy consumption.

The Critical Gap

Current Gaia-X compliance dimensions:

✓ Data sovereignty (GDPR compliance)
✓ Interoperability (API and data standards)
✓ Security (encryption, access control, identity)
✗ Energy footprint (UNMEASURED)
✗ Resource intensity (UNMEASURED)
✗ Thermodynamic efficiency (IGNORED)

As a result, digital infrastructures may be fully compliant with Gaia-X governance principles while remaining **opaque in terms of their physical energy demands**.

EFU-E (Energy-Flow Unit – Energy) addresses this gap by introducing a **standardized, human-scale metric** for reporting energy consumption alongside existing governance criteria.

EFU-E Integration Framework for Gaia-X Nodes

Illustrative Data Center Assessment

Operational Energy:

IT load: 10 MW (continuous)
Cooling overhead (PUE 1.5): 5 MW
Total operational load: 15 MW

Annual operational energy:

15 MW × 31,536,000 s/year = 473,040 MJ/year
473,040 MJ / 3,154 MJ per EFU-E = 150,000 EFU-E/year

Embodied Energy (Construction and Equipment)

Steel, concrete, servers: ~50,000 TJ (one-time)
Amortization period: 15 years
Annualized energy: 3,333 TJ/year = 3,333,000 MJ/year

Converted to EFU-E:

3,333,000 MJ / 3,154 = 1,057 EFU-E/year

Total Facility Energy Footprint

Operational EFU-E: 150,000
Embodied EFU-E: 1,057

Total: 151,057 EFU-E/year

(All conversions use 1 EFU-E = 3,154 MJ/year/person.)

Service Intensity and Human-Scale Interpretation

Annual data processed: 100 PB
Energy intensity: 151,057 EFU-E / 100 PB
= 1,511 EFU-E per PB

Population-Equivalent Interpretation

"This Gaia-X node consumes energy equivalent to the **annual metabolic energy of 151,057 people** in order to process 100 petabytes of data."

Such framing translates abstract infrastructure metrics into **intuitively comprehensible societal terms**, without replacing conventional energy units.

Benchmark Target (Indicative, 2030)

Target intensity: <500 EFU-E per PB
Required improvement: ~67% efficiency gain
Renewable share: >80%

Policy Recommendation: Gaia-X Certification Extension (Proposed)

Gaia-X Certification – Resource Accountability Clause (Draft)

"Gaia-X compliant nodes shall report annually:

(a) Total energy consumption expressed in EFU-E units
(b) Energy intensity per exabyte of data processed (EFU-E/EB)
(c) Share of renewable energy sourcing (%)
(d) Year-over-year efficiency improvement (% EFU-E reduction)

Methodology: EFU-Energy Standard v1.0 (Simor, 2026)
Verification: Independent audit required for nodes exceeding
10,000 EFU-E per year."

EFU-E reporting **extends existing sustainability disclosure practices** into the digital infrastructure domain, without introducing novel regulatory instruments.

3.2 AI Act and Computational Resource Governance

The EU Artificial Intelligence Act (2024) classifies AI systems according to risk (minimal, limited, high, unacceptable) and imposes obligations related to transparency, data governance, and human oversight. However, it does not address the **material and energetic costs** associated with large-scale computational systems.

The Materiality Blind Spot

High-Risk AI System (e.g. biometric identification):

Current AI Act obligations:
✓ Risk assessment documentation
✓ Data governance and quality controls
✓ Transparency and explainability
✓ Human oversight mechanisms
✗ Energy consumption disclosure (MISSING)
✗ Resource efficiency targets (MISSING)

As a consequence, two AI systems with identical functional performance may differ by **orders of magnitude in energy demand**, without this difference being visible to regulators or procurers.

EFU-E Contribution to AI Governance

Illustrative Example: GPT-4-Scale Model

Training Phase (one-time):

Compute load: ~25,000 GPU-days (NVIDIA A100 class)
Energy use: ~10,000 MWh = 36,000 GJ

Converted to EFU-E:

36,000 GJ = 36,000,000 MJ
36,000,000 / 3,154 = 11,410 EFU-E

Inference Phase (annual):

Average inference load: 6 MW (continuous)
Annual energy: 6 MW × 8,760 h = 52,560 MWh
= 189,216 GJ

Converted to EFU-E:

189,216 GJ = 189,216,000 MJ
189,216,000 / 3,154 = 59,993 EFU-E/year

Total Lifecycle Energy (Year 1):

Training: 11,410 EFU-E
Inference: 59,993 EFU-E

Total: 71,403 EFU-E

(All conversions use 1 EFU-E = 3,154 MJ/year/person.)

Interpretive Framing

"Deploying this AI system requires energy equivalent to the **annual metabolic consumption of 71,403 people**."

Per-Query Energy Intensity

Annual queries: 100 million/day = 36.5 billion/year
71,403 EFU-E / 36.5 × 10^9 queries
= 1.96 × 10^{-6} EFU-E per query

"Each query consumes energy equivalent to approximately **two millionths of a person's annual metabolism**."

EFU-E disclosure does **not impose hard energy caps**, but introduces **proportional transparency**, enabling differentiation between justified computational demand and avoidable excess.

Proposed Amendment to AI Act – Article 13 (Transparency)

Article 13(3)(f) – Proposed Addition:

"Providers of high-risk AI systems shall include in the technical documentation:

(f) Annual energy consumption calculated according to the EFU-Energy Standard v1.0 (Simor, 2026), including:

(i) Training phase energy (EFU-E, one-time)
(ii) Inference phase energy (EFU-E per year)
(iii) Energy per 1,000 transactions or queries (EFU-E/k)
(iv) Efficiency improvement targets (% annual reduction)
(v) Renewable energy share (%)

Reporting shall be public for systems deployed within the European Union. Systems exceeding 100,000 EFU-E per year shall require independent verification."

Implications for Public Procurement

Example: Public Sector AI Tender – Facial Recognition

Technical requirements:
- Accuracy: >99.5%
- False positive rate: <0.1%
- Throughput: 1,000 faces/hour
- Energy efficiency: <0.5 EFU-E per 1,000 scans (NEW)

Evaluation weights:
- Accuracy: 40%
- Security and privacy: 30%
- Cost: 20%
- Resource efficiency: 10% (EFU-E)

This framework incentivizes **algorithmic sufficiency** rather than unbounded computational scaling.

3.3 The "One-Armed Giant" Problem in European Digital Policy

Current Governance Landscape

Digital governance instruments:

✓ GDPR – privacy and data protection
✓ DSA – content moderation
✓ DMA – competition regulation
✓ AI Act – algorithmic risk management

Physical resource governance:

✗ Energy flows (unmeasured)
✗ Material flows (unmeasured)
✗ Thermodynamic efficiency (ignored)
✗ Infrastructure externalities (externalized)

The result is a structurally imbalanced system: **strong protection of digital rights paired with weak accountability for physical resource use**.

This condition can be described as a **"One-Armed Giant"**: powerful in normative governance, yet blind to its energetic and material foundations.

EFU-E as a Completing Layer

Integrated Governance Perspective:

Data layer (existing):
- Who controls data?
- How is it processed?
- What rights apply?

Resource layer (EFU-E):
- What is the energy cost?
- Where does the energy originate?
- How efficiently is it used?

Combined framing:

"This AI system processes biometric data in a GDPR-compliant manner, consumes 50,000 EFU-E annually, and is powered by 70% renewable energy."

Without EFU-E-type metrics, European digital governance **protects citizens in cyberspace while remaining blind to the physical costs imposed on the biosphere**.

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4. CONVERSION FORMULAS

4.1 Energy-to-EFU Conversion

Basic Formula

EFU-E = (MJ/year) / 3,154

Or from common units:

EFU-E = (GJ/year) / 3.154
EFU-E = (kWh/year) / 876
EFU-E = (kcal/year) / 730,000

Example

Household electricity: 10,000 kWh/year
EFU-E = 10,000 / 876 ≈ 11.4 EFU-E

Interpretation:
"This household uses energy equivalent to 11.4 people's metabolic consumption, just for electricity"

4.2 From Fuel Consumption

Common Fuels and EFU-E Conversion

Fuel	Energy Content	EFU-E per unit
Natural gas	38 MJ/m³	0.012 per m³
Gasoline	34.2 MJ/L	0.0108 per L
Diesel	38.6 MJ/L	0.0122 per L
Coal (bituminous)	24 MJ/kg	0.0076 per kg
Wood (dry)	18 MJ/kg	0.0057 per kg
Electricity	3.6 MJ/kWh	0.0011 per kWh

Vehicle Example

Car consumes 2,000 L gasoline/year
Energy: 2,000 × 34.2 = 68,400 MJ
EFU-E = 68,400 / 3,154 ≈ 21.7 EFU-E

Interpretation:
"This vehicle uses energy equivalent to 22 people's
annual metabolic consumption"

4.3 From Power (Continuous)

For Continuous Processes

EFU-E = (Power in W × 31,536,000 s/year) / 3,154,000,000
= Power in W / 100

Or simplified:

EFU-E/year = (kW continuous) × 8.76

Example: Data Center

Data center: 5 MW continuous
EFU-E = 5,000 kW × 8.76 ≈ 43,800 EFU-E/year

Interpretation:
"This data center consumes energy equivalent to 43,800 people's
annual metabolic energy"

4.4 EFU-E Conversion & Energy Flow Overview

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┌──────────────────────────────────┐
│      ENERGY INPUTS               │
│ (Fuels, Electricity, Power)      │
└──────────────────────────────────┘
          │ convert to EFU-E
          ▼
┌──────────────────────────────────┐
│    EFU-E UNITS STANDARD          │
│ Base: 1 EFU-E = 3,154 MJ/year    │
│      = 100 W continuous          │
│      = 2,000 kcal/day            │
│      = 876 kWh/year              │
└──────────────────────────────────┘
          │ interpret as
          ▼
┌──────────────────────────────────┐
│ HUMAN-EQUIVALENT ENERGY          │
│  (Metabolic perspective)         │
│ - Household: 10,000 kWh ≈ 11.4 EFU-E │
│ - Car: 2,000 L gasoline ≈ 21.7 EFU-E │
│ - Data center: 5 MW ≈ 43,800 EFU-E │
└──────────────────────────────────┘
          │ combine with
          ▼
┌──────────────────────────────────┐
│ ENERGY INVESTMENT & EROI         │
│ - Hydro 100:1 → 0.01 EFU-E input │
│ - Coal 50:1 → 0.02 EFU-E input   │
│ - Solar PV 10:1 → 0.10 EFU-E input │
│ Low EROI → energy trap risk      │
└──────────────────────────────────┘
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5. ECONOMIC & THERMODYNAMIC ANALYSIS

5.1 Energy Return on Investment (EROI)

Definition:
Ratio of energy delivered to energy invested in extraction or production.

EROI by Source

Energy Source	EROI	EFU-E input per EFU-E output
Hydropower	100:1	0.01
Nuclear	75:1	0.013
Coal	50:1	0.02
Wind	20:1	0.05

Energy Source	EROI	EFU-E input per EFU-E output
Oil (conventional)	20:1	0.05
Solar PV	10:1	0.10
Tar sands	3:1	0.33
Corn ethanol	1.3:1	0.77

Implications

Low EROI (e.g., tar sands at 3:1):
- To deliver 100 EFU-E net energy → must extract 133 EFU-E gross
- Investment: 33 EFU-E

High EROI (e.g., hydro at 100:1):
- To deliver 100 EFU-E net → extract 101 EFU-E gross
- Investment: 1 EFU-E

Key Insight: Energy transitions must maintain sufficient EROI.

- Too low → "energy trap" (more energy required to get energy than obtained).

Policy Threshold

Minimum viable EROI for complex society: 5:1
- Below 5:1 → Subsidy-dependent, not self-sustaining
- Corn ethanol (1.3:1) fails this test
- Tar sands (3:1) marginally viable only with subsidies

5.2 Jevons Paradox & Rebound Effects

Problem: Efficiency gains may increase total consumption.

Example: Personal Transportation

Baseline:
- Car efficiency: 8 L/100 km
- Annual driving: 15,000 km
- Fuel: 1,200 L = 41,040 MJ = 13.0 EFU-E

After efficiency improvement to 5 L/100 km

Expected savings (technical): 13.0 × (5/8) = 8.1 EFU-E

Actual savings (with rebound effects):
1. Direct rebound: Drive more (cheaper per km)
   - New distance: 18,000 km
   - Fuel: 900 L = 30,780 MJ = 9.8 EFU-E

2. Indirect rebound: Spend fuel savings on other energy
   - Example: Flight with saved money = 0.6 EFU-E

Total actual savings: 9.8 + 0.6 = 10.4 EFU-E
- Only ~20% reduction vs. 37.5% technical potential

Policy Implication

Efficiency improvements alone insufficient
Must combine with:
- Absolute caps (EFU-E budgets)
- Behavioral change
- Structural transformation

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6. IMPLEMENTATION GUIDELINES

6.1 Data Collection Requirements

Minimum Data (mandatory for EFU-E reporting):

1. **Annual energy consumption by fuel type** (MJ, kWh, L, m³)
2. **Energy conversion factors** (if non-standard fuels)
3. **System boundaries:** operational, embodied, scope 1/2/3
4. **Temporal boundaries:** annual standard

Recommended Data (enhances precision and comparability):

1. Monthly/seasonal energy variation
2. End-use breakdown (heating, cooling, processes, transport)
3. Conversion equipment efficiency
4. Renewable vs. fossil energy share
5. On-site generation (solar, CHP)

Advanced Data (for in-depth analysis):

1. Exergy flows (available work)
2. Cascade energy systems
3. Waste heat recovery
4. Lifecycle embodied energy

□□ Note: Minimum data is required for compliance; recommended and advanced data provide richer insight for policy and audit purposes.

6.2 Reporting Standards

Required Disclosure (Energy Impact Statement):

Energy Impact Statement

- Total energy consumption: [X] MJ/year = [Y] EFU-E
- Energy source breakdown: Fossil [%] / Renewable [%] / Nuclear [%]
- Reference baseline: 1 EFU-E = 3,154 MJ/year (human metabolic equivalent)
- Methodology: EFU-Energy Standard v1.0 (Simor, 2026)
- Scope: Operational / Operational + Embodied / Full Lifecycle
- Per capita (if applicable): [EFU-E per employee/resident]
- Data quality / uncertainty: [±X%]

Implementation Notes:

- EFU-E reporting **must specify system and temporal boundaries** to ensure comparability.
- **Primary vs. final energy** distinction must be explicit (see Ch.2).
- For **multi-site operations**, aggregate EFU-E per facility and per service unit.
- Optional: Visual dashboards can map EFU-E flows by energy type and end-use for internal and public communication.

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7. CASE STUDIES

7.1 Household Energy Footprint (Germany)

Profile: 4-person family, 120 m² apartment, well-insulated

Annual Energy Consumption:

Category	Consumption	EFU-E total	EFU-E per capita
Electricity	4,500 kWh	5.14	1.28
Heating (gas)	1,500 m³	18.07	4.52
Car (diesel)	900 L	11.01	2.75
Public transport	8,000 pass-km	0.38	0.10
Air travel (family)	4,000 km	20.32	5.08
Embodied (food, goods)	–	60	15.00
Total	–	114.92	28.7

Assessment:

- □ Below European average (40 EFU-E/person)
- □ Above sustainable target (15–20 EFU-E/person) by 44–91%
- Main contributors to excess: **Air travel (5 EFU-E)** + **embodied consumption**

Reduction Pathway to 20 EFU-E/person:

Measure	Reduction EFU-E/person
Eliminate air travel	5
Shift car to e-bike/train	2
Reduce meat consumption	1.5
Total Reduction	8.5 → 20.2 EFU-E/person ✓

7.2 Office Building Audit (Frankfurt)

Profile: 10,000 m² modern office, 500 employees

Annual Energy Consumption:

Category	Consumption	EFU-E total
Electricity	2,512,500 kWh	2,868
District heating	1,500 GJ	476
Total operational	–	3,344

Per Unit:

- Per m²: 0.334 EFU-E/m²/year (293 kWh/m²/year)
- Per employee: 6.69 EFU-E/person/year

Benchmark Comparison:

Standard	EFU-E/m²/year
Best practice (passive)	0.032
Modern standard	0.079
This building	0.334 (HIGH)
Older average	0.159

Improvement Scenario:

Measure	Savings EFU-E	% Reduction
LED retrofit + controls	428	30% lighting
HVAC optimization	171	20% reduction
Behavioral (shutdown protocols)	36	10% IT
District heating efficiency	71	15% reduction
Total savings	706	21% overall

Post-Improvement Totals:

- Total: 2,638 EFU-E/year
- Per employee: 5.28 EFU-E/person (21% reduction)

7.3 National Energy Transition ("Sustainia")

Baseline (2025):

Metric	Value
Population	50 million
Total energy	5,000 PJ = 1,585 billion EFU-E
Per capita energy	31.7 EFU-E/person
Fossil share	90% (1,427 billion EFU-E)
Per capita fossil	28.5 EFU-E/person

Target (2050):

Metric	Value
Population	55 million
Total energy	3,500 PJ = 1,110 billion EFU-E (30% reduction)
Per capita energy	22.2 EFU-E/person
Fossil share	20% (222 billion EFU-E, −84%)
Per capita fossil	4.0 EFU-E/person (−86%)
Renewable share	55%

Investment & Returns:

Category	Value
Total investment	$500 billion over 25 years = 1.3% GDP annually
Energy cost savings	475 billion EFU-E avoided
Monetary equivalent	$7.1 trillion saved at $15/EFU-E
Net benefit	$6.6 trillion (13:1 ROI)
CO_2 reduction	6 Gt cumulative ($50/tonne → $300 billion carbon value)
Additional benefits	Air quality, energy security, job creation, tech export

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8. APPENDICES

APPENDIX A: QUICK REFERENCE CARD

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║                EFU-Energy Standard v1.0                       ╖
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BASELINE: 1 EFU-E = 3,154 MJ/year/person
          = 100 W continuous
          = 876 kWh/year
          = 2,000 kcal/day

CONVERSIONS:
• GJ/year → EFU-E: Divide by 3.154
• kWh/year → EFU-E: Divide by 876
• W continuous → EFU-E/year: Divide by 100
• L gasoline → 0.0108 EFU-E
• m³ natural gas → 0.012 EFU-E

INTERPRETATION:
100 EFU-E = "Energy flow equivalent to metabolic
        consumption of 100 people"

BENCHMARKS:
• Energy poverty: <3 EFU-E/person
• Decent living: 6-10 EFU-E/person
• Sustainable target: 15-20 EFU-E/person
• Global average: 23.8 EFU-E/person
• High consumption: >40 EFU-E/person

KEY RELATIONSHIPS:
• 1 EFU-E (coal) ≈ 1.13 EFU-C (carbon)
• 1 EFU-E (fossil) ≈ 0.6-0.7 EFU-S (entropy)
• Primary vs. Final: Specify which used

CRITICAL RULE:
Fossil fuels → Report PRIMARY energy
Renewables → Report FINAL energy
(Ensures thermodynamic honesty)

CITATION:
Simor, I. (2026). EFU-Energy Standard v1.0.
Nookratic Bloc. CC BY 4.0.
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APPENDIX B: FUEL CONVERSION TABLE

Common Fuels – Detailed Reference (Net Calorific Value)

Fuel Type	Unit	Energy Content (MJ)	EFU-E per unit
Solid Fuels			
Anthracite coal	kg	27	0.00856
Bituminous coal	kg	24	0.00761
Lignite coal	kg	15	0.00476
Charcoal	kg	30	0.00951

Fuel Type	Unit	Energy Content (MJ)	EFU-E per unit
Wood (dry, 20% moisture)	kg	15	0.00476
Wood pellets	kg	17	0.00539
Peat	kg	12	0.00380
Liquid Fuels			
Crude oil	L	38	0.01205
Gasoline	L	34.2	0.01084
Diesel	L	38.6	0.01224
Jet fuel (kerosene)	L	37	0.01173
Heavy fuel oil	L	40	0.01268
LPG (propane)	L	25.3	0.00802
Ethanol	L	23.4	0.00742
Biodiesel	L	33	0.01046
Methanol	L	15.6	0.00495
Gaseous Fuels			
Natural gas	m^3	38	0.01205
LNG (liquefied)	kg	55	0.01744
Propane	m^3	93.3	0.02958
Butane	m^3	122	0.03868
Hydrogen	m^3	12.7	0.00403
Hydrogen	kg	142	0.04502
Biogas (60% CH_4)	m^3	21.5	0.00682
Other			
Electricity	kWh	3.6	0.00114
District heating	GJ	1,000	0.317

Notes:

- All values at standard conditions (STP for gases: 0°C, 1 atm)
- Wood moisture content significantly affects energy value
- Natural gas composition varies by region (±10%)
- LPG may be propane, butane, or mixture

APPENDIX C: SECTORAL ENERGY INTENSITY BENCHMARKS

Manufacturing (EFU-E per tonne product)

Product	Best Practice	Industry Average	High Intensity
Steel (primary)	6.3	9.5	15.8
Steel (recycled)	1.9	3.2	4.7
Aluminum (primary)	47.5	63.3	79.1
Aluminum (recycled)	3.2	4.7	6.3
Cement	2.5	3.8	5.7
Glass	4.1	6.3	9.5
Plastics (polyethylene)	19.0	25.3	31.6
Paper (virgin)	7.6	12.7	19.0
Paper (recycled)	3.2	6.3	9.5
Ammonia (Haber-Bosch)	9.5	11.4	14.2
Textiles (cotton fabric)	31.6	47.5	63.3
Semiconductors (per wafer)	158.5	253.6	380.4

Services (EFU-E per $1,000 revenue)

Sector	Best Practice	Industry Average
Banking/Finance	0.5	1.2
Insurance	0.6	1.5

Sector	Best Practice	Industry Average
Legal services	0.8	2.0
Software/IT	1.5	3.5
Retail (general)	2.5	5.0
Grocery retail	4.0	7.0
Hospitality (hotels)	12.0	20.0
Restaurants	8.0	15.0
Healthcare (hospitals)	15.0	25.0
Education (universities)	8.0	12.0

Transport (EFU-E per passenger-km or tonne-km)

Mode	Passenger (per 100 km)	Freight (per 1,000 tonne-km)
Walking	0.0048 (metabolic)	N/A
Bicycle	0.0016 (metabolic)	N/A
E-bike	0.0032	N/A
Motorcycle	0.019	N/A
Car (gasoline, solo)	0.063	N/A
Car (EV, solo)	0.019	N/A
Bus (diesel)	0.0095	N/A
Bus (electric)	0.0063	N/A
Train (electric)	0.0048	9.5
Tram/Metro	0.0063	N/A
Ferry	0.025	25.3
Domestic flight	0.19	47.5
Long-haul flight	0.127	31.6

Mode	Passenger (per 100 km)	Freight (per 1,000 tonne-km)
Truck (diesel)	N/A	31.6
Truck (electric)	N/A	12.7
Rail freight	N/A	6.3
Ship (container)	N/A	3.2
Pipeline (oil/gas)	N/A	0.95

APPENDIX D: GLOSSARY

Primary Energy: Energy in natural form before conversion (coal in ground, crude oil, flowing water, wind, sunlight).

Final Energy: Energy delivered to the end user (electricity at socket, gasoline at pump, district heat at building).

Useful Energy: Energy that actually performs desired service (light, motion, heat). Always less than final energy due to end-use losses.

Energy Intensity: Energy per unit of output (EFU-E per GDP, per tonne product, per m² building).

Exergy: Available work content of energy; quality measure. High exergy = electricity, low exergy = waste heat.

EROI (Energy Return on Investment): Ratio of energy delivered to energy invested in extraction/production.

Embodied Energy: Energy consumed in manufacturing, transport, and construction of goods (excludes operational energy).

Operational Energy: Energy consumed during product/building use phase.

Primary Energy Equivalent: Method of accounting electricity at thermal equivalent for comparability with fossil fuels.

Capacity Factor: Actual energy output divided by theoretical maximum (full capacity 24/7).

Rebound Effect: Efficiency gains leading to increased consumption, partially offsetting savings.

Energy Poverty: Insufficient access to energy for basic needs (cooking, lighting, heating/cooling, communication).

Energy Justice: Fair distribution of energy benefits and burdens.

Decoupling: Breaking the link between economic growth and energy/carbon growth (relative or absolute).

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9. VALIDATION & UNCERTAINTY

9.1 Reference Value Variability

Human Energy Expenditure Ranges:

Category	Energy (MJ/year)	EFU-E
Infant (1 year)	1,460	0.46
Child (10 years)	2,190	0.69
Adult female (sedentary)	2,555	0.81
Adult male (sedentary)	2,920	0.93
Adult (average activity)	3,154	1.00
Adult (active)	3,650	1.16
Athlete (training)	5,475	1.74

Adopted Standard:

1 EFU-E = 3,154 MJ/year (100 W continuous, average adult)
Uncertainty: ±15% (activity level, body mass variation)

9.2 Measurement Uncertainty

Typical Uncertainties:

Source	Uncertainty
Utility meter readings	±2%
Fuel consumption (fleet)	±5–10%
Embodied energy (LCA)	±30–50%
Grid electricity carbon intensity	±10–20%
Efficiency estimates	±10–15%

Reporting Requirement:

All EFU-E values must include uncertainty.
Example: 50,000 EFU-E (±8%)

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10. POLICY APPLICATIONS

10.1 Corporate Energy Reduction Scheme

Mandate: Large energy users (>10,000 EFU-E/year) must:

1. Report annual EFU-E consumption
2. Achieve 2% reduction per year (2025–2030)
3. 1% reduction per year (2030–2040)
4. Maintain absolute levels (2040–2050) despite growth

Example:

Year	Target EFU-E
2025	100,000
2030	90,392
2040	81,850
2050	81,850

If company grows revenue 50% (2025–2050): Energy intensity must fall −63%

10.2 Residential Energy Performance Standards

New Buildings (from 2026):

Maximum: 5 EFU-E per occupant/year (~1,575 kWh/person/year electricity equivalent)

Existing Buildings (retrofit by 2040):

Maximum: 12 EFU-E per occupant/year

Incentives:

- Buildings <3 EFU-E/person: Tax credit
- Buildings >20 EFU-E/person: Energy tax

10.3 Personal Energy Allowances (PEA)

Concept: Cap-and-trade for individuals

Allocation: 15 EFU-E/year per citizen (tradable)

Example:

User Type	Consumption	Action	Cost/Income
High consumer	40 EFU-E	Buy 25 EFU-E permits	$250
Low consumer	8 EFU-E	Sell 7 EFU-E permits	$70

Benefits:

- Rewards efficiency financially
- Penalizes high consumption
- Revenue-neutral (permits circulate)
- Declining cap drives reduction over time

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11. AUDIT PROTOCOL

11.1 Verification Checklist

Level 1: Self-Assessment

- ☐ Energy bills/meter readings collected
- ☐ Fuel consumption records compiled
- ☐ Conversion factors applied correctly
- ☐ System boundaries defined
- ☐ Primary vs. final energy specified

Level 2: Third-Party Audit

- ☐ Utility data independently verified
- ☐ Fuel delivery records cross-checked
- ☐ On-site generation validated (solar/CHP)
- ☐ Scope 1/2/3 boundaries confirmed
- ☐ Calculation methodology reviewed

Level 3: Certification (ISO 50001 aligned)

- ☐ Energy management system in place
- ☐ Sub-metering for major loads
- ☐ Energy performance indicators tracked

- ☐ Improvement targets set and monitored
- ☐ Multi-year trend analysis

11.2 Common Errors

Mistake	Wrong Example	Correct Approach
Double-counting electricity	Count both kWh and fuel at power plant	Count electricity at point of use (primary equivalent included)
Ignoring on-site generation	Report only purchased energy	Subtract on-site solar/wind/CHP; e.g., 60,000 kWh purchased − 10,000 kWh solar = 50,000 kWh net
Mixing energy and power	"Building uses 500 kW"	Convert to energy: 500 kW × 8,760 h = 4,380 MWh/year = 5,000 EFU-E
Wrong unit conversion	1 MWh = 1,000 MJ	1 MWh = 3,600 MJ = 3.6 GJ = 1.14 EFU-E
Forgetting embodied energy	"Electric car uses zero energy"	Include manufacturing: 79 EFU-E, operation: 69 EFU-E over 10 years

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12. INTEGRATION WITH OTHER EFU STANDARDS

12.1 EFU-Carbon Linkage

Stoichiometric Relationship:

Combustion: Fuel + $O_2 \rightarrow CO_2 + H_2O$

Mass Ratios (approximate):

- 1 kg CO_2 emitted ≈ 1 kg O_2 consumed (fossil fuels)

EFU Relationship:

Energy Source	1 EFU-E → EFU-C
Coal	1.13 EFU-C
Natural gas	0.67 EFU-C
Oil	0.88 EFU-C

Energy Source	1 EFU-E → EFU-C
Renewables/Nuclear	0 EFU-C

Reference Values:

- 1 EFU-C = 263 kg CO_2/year
- O_2 consumed ≈ 1.14 EFU-O_2/year

12.2 Joint Reporting Example

Company Fossil Fuel Use:

Metric	Value
CO_2 emissions	100,000 t/year = 380,228 EFU-C
O_2 consumption	~100,000 t/year = 434,783 EFU-O_2
Energy (coal, primary)	1,064 TJ = 337,350 EFU-E

Validation Ratio: 1.13 EFU-C per EFU-E (coal) ✓

12.2 EFU-Entropy Linkage

Energy Degradation:

All energy use generates entropy (see EFU-Entropy Standard)

Example: Thermal Plant (~35% efficiency)

Metric	Value
EFU-E consumed	1.00
Useful work	0.35 EFU-E
Waste heat	0.65 EFU-E → entropy generation
Entropy (J/K)	7,116
EFU-S	0.619

Rule of Thumb:

1 EFU-E (fossil) ≈ 0.6–0.7 EFU-S

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13. FUTURE DEVELOPMENTS

13.1 Planned Enhancements (2026–2027)

1. Sectoral Benchmarking Database

- Industry-specific EFU-E/unit production norms
- Best-practice vs. average performance
- Real-time updates

2. Embodied Energy Calculator

- Product lifecycle EFU-E footprints
- Materials database (steel, concrete, plastics)
- Manufacturing process energy

3. Dynamic Grid Integration

- Real-time grid carbon/energy intensity
- Time-of-use EFU-E accounting
- Renewable % in electricity mix

4. Personal Energy Tracking

- Consumer app (scan purchases → EFU-E)
- Gamification of reduction targets
- Social comparison (anonymized)

13.2 Research Priorities

1. Energy–Wellbeing Correlation

Question: What is minimum EFU-E for maximum wellbeing?
Evidence: Diminishing returns above ~20 EFU-E/person
High consumers (>60 EFU-E) not significantly happier
Research needed: Cultural contexts, climate zones

2. Just Transition Pathways

How to reduce high-consumer EFU-E without social disruption?
How to increase low-consumer EFU-E affordably?
Role of technology vs. behavior change

3. Breakthrough Technologies

Fusion energy: Potential 1,000:1 EROI → impact on EFU-E accounting
Direct air capture: Energy cost per tonne CO_2
Advanced batteries: Embodied EFU-E trends

4. Energy Democracy

Community energy: Distributed EFU-E generation
Prosumer models: Self-sufficiency potential
Energy sovereignty: Reducing dependence

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14. CONCLUSIONS FOR EDUARDO / FUTURIBLES

14.1 The EFU-E Contribution to European Foresight

1. **Materiality of the Digital**
   The EFU-E framework highlights the **physical resource basis of digital infrastructure**, currently unaccounted for in Gaia-X and the AI Act. Every exabyte stored, every AI model trained, and every blockchain transaction consumes **energy that can be expressed in human metabolic equivalents**, making the invisible visible.
2. **Beyond Carbon Tunnel Vision**
   While carbon-focused frameworks (EFU-C) capture climate impacts, EFU-E measures **resource depletion, thermodynamic efficiency, and energy justice simultaneously**. This multi-dimensional perspective is essential for **holistic energy and digital governance**.
3. **The "One-Armed Giant" Problem**
   European digital policy (GDPR, DSA, DMA, AI Act) excels in regulating **data flows**, yet remains blind to **energy and material flows**. EFU-E closes this gap, enabling:

- □ Resource-conscious AI procurement
- □ Data center sustainability benchmarking
- □ Digital sufficiency policies (beyond mere efficiency)
- □ True cost accounting for cloud and AI services

4. **Narrative Power**
   EFU-E converts abstract energy units (Joules, MWh) into **relatable human-scale equivalents**. This narrative strength supports:

- Public engagement ("This AI system uses the energy of 71,000 people annually")
- Policy and political will-building (tangible, relatable targets)
- Democratic accountability (citizens can grasp the stakes of digital infrastructure)

14.2 Strategic Recommendations for EU Policy

Short-Term (2026–2027):

1. Implement voluntary **EFU-E reporting in 10 Gaia-X nodes**
2. Amend **AI Act Article 13** to include mandatory energy disclosure
3. Launch the **EU Energy Intensity Database** for sectoral benchmarks

Medium-Term (2028–2030):

1. Make **EFU-E reporting mandatory for all high-risk AI systems**
2. Require **Gaia-X nodes to achieve <500 EFU-E/PB by 2030**

3. Introduce **public procurement criteria with 10% weight on energy intensity**

Long-Term (2031+):

1. Introduce a **Digital Resource Cap**: EU-wide limit on data center EFU-E growth
2. Issue **Energy Passports for digital services** (analogous to building EPCs)
3. Establish an **International EFU-E Technical Standard** under ISO guidance

These measures ensure that **digital sovereignty is complemented by physical accountability**, aligning Europe's AI and cloud strategies with sustainability objectives.

14.3 Article Outline for Futuribles

Suggested Title:
"Beyond Data Sovereignty: The Missing Materiality of Europe's Digital Future"

Proposed Structure:

1. **The Paradox:** Europe excels in digital rights but lags in resource governance
2. **Case Study:** Gaia-X node energy audit (fictional but illustrative)
3. **The EFU-E Solution:** Human-centered metrics for energy and resource flows
4. **Policy Proposal:** Integration of EFU-E into AI Act and Gaia-X certification
5. **Vision 2050:** EU as the first jurisdiction governing both digital and physical infrastructure holistically

Key Messages:

- Digital sovereignty is incomplete without resource accounting
- EFU-E provides the missing **physical layer** for informed governance
- Europe has the opportunity to **lead globally by example**
- **Time to act:** Mitigate AI and cloud energy footprints before they become unmanageable

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16. CONTACT & CONTRIBUTIONS

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How to Contribute

1. Submit feedback via GitHub: nookratikus-blokk/efu-energy
2. Propose methodology improvements
3. Share implementation case studies
4. Translate to additional languages
5. Develop calculation tools and apps

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17. LICENCE

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VERSION HISTORY

Version	Date	Changes
1.0	2026-01-01	Initial release with Gaia-X/AI Act integration

Planned Updates

- **v1.1 (Q2 2026):** Add sectoral benchmark database
- **v1.2 (Q4 2026):** Integrate dynamic grid carbon intensity
- **v2.0 (2027):** Exergy analysis module

Glossary (EFU-E Energy Standard v1.0)

EFU-E (Equivalent Flux Unit – Energy)
Unit measuring energy consumption in **human metabolic equivalents**.

- **1 EFU-E = 3,154 MJ/year/person**
- **= 100 W continuous metabolic power**
- **= 2,000 kcal/day/person**
- **= 876 kWh/year/person**

Basal Metabolic Rate (BMR)
Energy required to sustain a resting adult's basic physiological functions (70–85 W).

Primary Energy
Total energy content of fuel **at the point of extraction**, before conversion losses. Used for fossil fuels in EFU-E accounting.

Final Energy
Energy delivered to the end-use device (electricity, heat, mechanical work). Used for renewables and nuclear in EFU-E accounting.

Personal Energy Allowance (PEA)
Per-capita energy budget aligned with **sustainable pathways** (e.g., 15–20 EFU-E/person/year by 2050).

Exergy
The portion of energy that can perform **useful work**. EFU-E reports gross energy by default; exergy can be specified separately.

Gaia-X Node
Federated, European cloud infrastructure unit. EFU-E reporting quantifies **operational and embodied energy**.

AI Act Article 13
EU regulation on transparency obligations for high-risk AI systems. EFU-E reporting integrates **resource and energy metrics**.

"One-Armed Giant"
Metaphor describing regimes that are strong on **data governance** but weak on **resource and energy governance**.

Illustrative Example / Disclaimer
Numerical examples (e.g., GPT-4 scale, Gaia-X node, Sustainia) are **for illustration only, not audited data**.

EFU-E Reporting Checklist (for Gaia-X nodes & AI Systems)

1. General Information

- ☐ Entity Name / Node ID / AI System Name
- ☐ Reporting Period (year)
- ☐ Reference Standard: EFU-Energy Standard v1.0 (Simor, 2026)
- ☐ Contact / Responsible Auditor

2. Energy Accounting

- ☐ Total **Primary Energy** (MJ & EFU-E) – fossil sources only
- ☐ Total **Final Energy** (MJ & EFU-E) – renewables & nuclear
- ☐ Operational vs. Embodied Energy (construction, servers, infrastructure)
- ☐ Continuous power load (W or MW)

3. Efficiency & Intensity Metrics

- ☐ Energy per data processed (EFU-E/PB or EFU-E/1,000 queries)
- ☐ Renewable energy share (%)
- ☐ Year-over-year efficiency improvements (% EFU-E reduction)

4. Compliance & Verification

- ☒ Independent audit conducted (if >10,000 EFU-E/year)
- ☒ Methodology documented
- ☒ Reporting aligns with EFU-E standard definitions
- ☒ Disclaimer for illustrative or modeled data included

5. Policy Alignment

- ☒ AI Act Article 13 requirements met
- ☒ Gaia-X certification thresholds met (<500 EFU-E/PB by 2030)
- ☒ Personal Energy Allowances (PEA) considered for sustainability benchmarking

6. Optional / Narrative Section

- ☒ Narrative translation of energy footprint into human-equivalents (e.g., “Energy of X people”)
- ☒ Contextual comparison to global/regional benchmarks
- ☒ Highlight areas for potential efficiency gains

END OF EFU-ENERGY STANDARD v1.0

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